

Aditi Hande

682-390-9300 | aditi.n.hande@gmail.com | github.com/adiHande | linkedin.com/in/aditi-hande | Arlington, TX

Education

University of Texas at Arlington

B.S. in Computer Science; GPA: 3.69

Honors: President's List, Honor Roll (2 semesters), Freshman Distinction Roll, ALD

Arlington, TX

Aug. 2023 – May 2027

Relevant Coursework

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Organization & Assembly, Databases, Operating Systems, Artificial Intelligence, Computer Networks, Information Security, Software Project Management, Data Mining, Machine Learning, Operations Research

Skills

Programming Languages: Python, C, C++, Java, MySQL, HTML, CSS

Tools & Platforms: Git/GitHub, Linux (Ubuntu, Fedora, WSL), MATLAB, MySQL, Roboflow, GIMP, Microsoft 365 Suite

Experience

UT Arlington Research Institute (UTARI)

Fort Worth, TX

Undergraduate Research Assistant

July 2026 – August 2026

- Evaluated AI-assisted annotation and object-detection workflows—including CLIP, Grounding DINO, and YOLO-based approaches—by comparing their performance and suitability for aerial imagery.
- Used CLIP for image-level analysis and Grounding DINO for prompt-based bounding-box generation in MATLAB; developed and modified class-specific text prompts to improve the model's detection performance.
- Manually corrected missing or inaccurate annotations and applied preprocessing and augmentation techniques in Roboflow to prepare aerial datasets for object-detection model training.
- Trained, validated, and tested YOLO models and assisted team members with laboratory and outdoor testing of a Raspberry Pi-based hog-feeder prototype.

U.S. Department of Agriculture, Agricultural Research Service (USDA-ARS)

Arlington, TX

Research Assistant Intern, USDA-ARS × UTA Summer Research Program

May 2026 – July 2026

- Worked on computer vision and machine learning models for automated wild hog detection in agricultural fields
- Implemented and evaluated YOLO object detection models in PyTorch, and fine-tuned hyperparameters to improve performance.
- Developed a repeatable, parametric synthetic farm dataset using hog, coyote, and swan decoys; with varied animal orientations, distances, backgrounds, lighting, and weather conditions and with fixed camera setup to identify conditions producing false positives and false negatives in YOLO models' detections.
- Built Python image-preprocessing and augmentation workflows using rembg, Segment Anything (SAM), depth maps, background replacement, and simulated weather effects to diversify object orientation, occlusion, lighting, and scene conditions.

University of Texas at Arlington, College of Engineering

Arlington, TX

Undergraduate Teaching Assistant, UNIV-EN 1131

Aug. 2025 – Dec. 2025

- Supported 150+ students across two sections by answering questions, clarifying concepts, guiding them through multi-session labs, and troubleshooting hands-on equipment
- Graded 15+ assignments per student and directed students to appropriate campus resources
- Assisted faculty with attendance tracking, equipment setup, and classroom/lab operations using Microsoft Forms, Excel, and Canvas a+ Attendance tools

Projects

Movie Theater System | *Python*

- Built a terminal-based booking system with movie selection, seat reservation, ticket pricing, and formatted ticket generation.

Restaurant Ordering System | *Python, MySQL*

- Built a terminal-based ordering system with categorized menus, multi-item ordering, tax calculation, and formatted receipt generation, integrating a MySQL database to store and retrieve transaction records

Mini Shell Program | *C, Linux*

- Built a partial Linux shell supporting file editing, process management, directory navigation, command execution, and file sorting operations

Airline Seat Reservation System | *C*

- Developed a reservation system using file handling, supporting multiple flight layouts, daily routes, seat availability tracking, and reservation logic based on flight ticket details

Bunny RPG Combat Game | *Java*

- Collaborated on a terminal-based RPG applying OOP principles (inheritance, polymorphism) to implement combat logic, a shop system, and ASCII-based UI